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A randomized controlled trial contrasting the effects of 4 low-calorie sweeteners and sucrose on body weight in adults with overweight or obesity

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ABSTRACT

Background: Low-calorie sweeteners (LCSs) provide sweetness with little or no energy. However, each LCS's unique chemical structure has potential to elicit different sensory, physiological, and behavioral responses that affect body weight.

Objective: The purpose of this trial was to compare the effects of consumption of 4 LCSs and sucrose on body weight, ingestive behaviors, and glucose tolerance over a 12-wk intervention in adults (18–60 y old) with overweight or obesity (body mass index 25–40 kg/m^2).

Methods: In a parallel-arm design, 154 participants were randomly assigned to consume 1.25–1.75 L of beverage sweetened with sucrose ($n = 39$), aspartame ($n = 30$), saccharin ($n = 29$), sucralose ($n = 28$), or rebaudioside A (rebA) ($n = 28$) daily for 12 wk. The beverages contained 400–560 kcal/d (sucrose treatments) or <5 kcal/d (LCS treatments). Anthropometric indexes, energy intake, energy expenditure, appetite, and glucose tolerance were measured at baseline. Body weight was measured every 2 wk with energy intake, expenditure, and appetite assessed every 4 wk. Twenty-four-hour urine collections were completed every 4 wk to determine study compliance via para-aminobenzoic acid excretion.

Results: Of the participants enrolled in the trial, 123 completed the 12-wk intervention. Sucrose and saccharin consumption led to increased body weight across the 12-wk intervention (Δweight $= +1.85 \pm 0.36$ kg and $+1.18 \pm 0.36$ kg, respectively; $P \leq 0.02$) and did not differ from each other. There was no significant change in body weight with consumption of the other LCS treatments compared with baseline, but change in body weight for sucralose was negative and significantly lower compared with all other LCSs at week 12 (weight difference $\geq 1.37 \pm 0.52$ kg, $P \leq 0.008$). Energy intake decreased with sucralose consumption ($P = 0.02$) and ingestive frequency was lower for sucralose than for saccharin ($P = 0.045$). Glucose tolerance was not significantly affected by any of the sweetener treatments.

Conclusions: Sucrose and saccharin consumption significantly increase body weight compared with aspartame, rebA, and sucralose, whereas weight change was directionally negative and lower for sucralose compared with saccharin, aspartame, and rebA consumption. LCSs should be categorized as distinct entities because of their differing effects on body weight. This trial was registered at clinicaltrials.gov as NCT02928653. *Am J Clin Nutr* 2019;109:1288–1301.

Keywords: low-calorie sweeteners, sweetness, ingestive behavior, energy intake, appetite, glycemia, overweight, obesity

Introduction

Low-calorie sweeteners (LCSs) are ubiquitous in the US food supply. They are consumed most commonly in beverages (by 30.8% of American adults), but also in foods (10.3%), and added

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Abbreviations used: ace-K, acesulfame potassium; ASA24, Automated Self-Administered 24-hour Dietary Recall; CPIR, cephalic phase insulin response; DTE, desire to eat; DXA, dual-energy X-ray absorptiometry; EE, energy expenditure; FFM, fat-free mass; FMC, fine motor control; GLP-1, glucagon-like peptide-1; HbA1c, glycated hemoglobin; iAUC, incremental AUC; LCS, low-calorie sweetener; mBEV, Beverage Consumption Habits Questionnaire; OGTT, oral-glucose-tolerance test; PABA, para-aminobenzoic acid; RCT, randomized control trial; RebA, rebaudioside A; SSB, sugar-sweetened beverage; T1R2/T1R3, taste 1 receptor member 2 / taste 1 receptor member 3 G-protein coupled heterodimer sweet taste receptor; TG, triacylglycerol; VAS, visual analog scale.

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to products via LCS packets (14.1%) (1). Results from meta-analyses of randomized controlled trials (RCTs) (2–4) conclude there is beneficial or no effect of LCS consumption on body weight compared with sugar-sweetened and water alternatives. Yet results from select epidemiological (5), in vitro (6), and animal trials (7) continue to raise consumer concerns about the efficacy of LCSs for weight management and glucose tolerance. Clinical and policy recommendations generally aggregate LCSs. This reflects a view that their common mode of action of LSCs is to provide sweetness while contributing little or no energy to the diet. However, each LCS has a unique chemical structure that elicits different sensory properties and digestibility, which may differentially affect appetite, brain reward activation, intestinal microbiota populations, metabolism, and body weight. If the latter is true, it may explain the variability in responses to single or combinations of LCSs in the literature.

At the sensory level, LCSs bind to different regions of the taste 1 receptor member 2 / taste 1 receptor member 3 (T1R2/T1R3) G-protein coupled heterodimer sweet taste receptor (8–10). This results in differences in cellular biophysical processes (11) that likely contribute to their unique sensory profiles (12), preingestive responses [e.g., cephalic phase insulin response (CPIR)], and downstream effects. A CPIR has been documented for saccharin (13) and potentially sucralose (14, 15), but not for other LCSs (16, 17). There is speculation that the CPIR can be disrupted with intermittent consumption of nutritive sweeteners and LCSs (18), uncoupling the association between taste and energy, and contributing to increased energy intake (19).

Although the T1R2/T1R3 receptor was originally identified in the oral cavity, this receptor is present throughout the gastrointestinal tract, pancreas, adipose tissue, hypothalamus, and elsewhere. Activation of receptors on cells in these tissues elicits responses characteristic of the tissue, raising new questions about whether LCSs are inert after swallowing. Differences in LCS digestibility can also contribute to disparate physiological outcomes. Aspartame, a dipeptide, is hydrolyzed to aspartate, phenylalanine, and methanol in the proximal intestinal lumen and has no documented effects on gastric emptying, gut peptide release, or glucose absorption (20). Steviol glycosides also undergo metabolism to glucuronide in the liver. Other LCSs are partially digested (~2.6% of sucralose) or remain intact [saccharin and acesulfame-potassium (ace-K)] (21, 22). These fates result in different opportunities to bind to the T1R2/T1R3 receptors on enteroendocrine cells and stimulate release of the gut peptides glucagon-like peptide-1 (GLP-1) and gastric inhibitory polypeptide; such has been reported with sucralose (23). Select LCSs are absorbed into the blood (saccharin, ace-K, a fraction of sucralose, and steviol) and are excreted in the urine (22). These sweeteners may interact with sweet receptors in the brain and periphery. Pancreatic β-cell exposure to saccharin, sucralose, and steviol elicits insulin release in vitro (24, 25) but such responses are not consistent across trials (26). If true, a LCS-mediated insulin release could increase hunger and energy intake (27). The postingestive consequences of different LCSs on reward-driven feeding remain unclear as well. Sweetness is a pleasant sensation; LCS and carbohydrate sweetener exposures differentially recruit food reward–related brain regions in both animals (28) and humans (29). Despite suggestions that sugar and LCSs promote cravings or addictive-like behavior (30), the available evidence is limited and these claims have not been substantiated (31). Some LCSs are excreted in the feces (sucralose, steviol glycosides, and a fraction of saccharin), with opportunities to alter the microbiota, metabolite production, or energy harvesting efficiency (22).

Before further investigation into the mechanisms by which LCSs may influence body weight, it is prudent to first establish that they do lead to differential weight outcomes when chronically consumed. Therefore, we conducted a 5-parallel-arm RCT to compare the effects of consumption of 4 commonly consumed LCSs [aspartame, saccharin, rebaudioside A (rebA), and sucralose] and sucrose on body weight, body composition, energy intake, energy expenditure (EE), appetite, and glycemia over 12 wk in adults with overweight or obesity [BMI (in kg/m^2) 25–40]. We hypothesized that the unique profile of physiological responses each LCS elicits would result in differential changes in body weight both between the tested LCSs and between LCS and sucrose.

Methods

Study overview

In a parallel-arm design, 154 participants were randomly assigned to consume beverages sweetened with 1 of 5 sweeteners (sucrose, saccharin, aspartame, rebA, or sucralose) daily for 12 wk. Saccharin, aspartame, rebA, and sucralose were selected for study, because they are the most widely consumed LCSs in the United States. Primary outcomes measured in this trial (NCT02928653) were body weight, body composition, dietary intake, EE, appetite, and glycemia. Secondary outcomes were serum lipids, sweetness perception, and overall liking of the delivery beverage. The urinary biomarker para-aminobenzoic acid (PABA) was used to monitor participant compliance with the beverage intervention. Participants were informed the study was an investigation of the effects of sweetener consumption on fine motor control (FMC). This was intended to minimize participant modification of diet or lifestyle that could affect body weight outside of the sweetener intervention. The research protocol was reviewed and approved by Purdue University's Institutional Review Board (#1510016667).

Participants

Participants were recruited from the Greater Lafayette, Indiana area from January, 2016 to March, 2018. Two hundred and ninety-three individuals completed the screening process and qualified for the trial. One hundred and fifty-four participants completed baseline procedures and were randomly assigned to a sweetener treatment. One hundred and twenty-three participants completed the entire 12-wk trial (see the Consolidated Standards of Reporting Trials flow diagram in **Supplemental Figure 1**). Participants completed a prescreening questionnaire eliciting health and lifestyle information and had height and weight measured. Eligibility criteria included *1*) 18–60 y of age, *2*) BMI between 25 and 40 kg/m^2, *3*) healthy (i.e., no history of diabetes, hypertension, etc.), *4*) low consumers of LCSs (reported consumption ≤1 time/wk), 5) weight stable (weight change ≤3

kg in the past 3 mo), and 6) Three Factor Eating Questionnaire dietary restraint score <14.

Protocol

Participants completed a series of 5 testing days at baseline. Three of these days included measurements of baseline energy intake and EE. On 1 of the 3 d, participants completed an appetite log for one 24-h period and one 24-h urine collection. The latter was to obtain a baseline amount of the urinary biomarker PABA to track compliance throughout the study. On the fourth baseline day, a dual-energy X-ray absorptiometry (DXA) scan and an oral-glucose-tolerance test (OGTT) were completed. On the final baseline day, participants reported for FMC testing, body weight measurements, and beverage sensory/hedonic evaluations. Measurements of FMC included hand steadiness, rotary tracking, and a questionnaire regarding FMC in daily life. Body weight and beverage sensory/hedonic evaluations were measured every 2 wk; assessment of energy intake, EE, and appetite and a 24-h urine collection were repeated during weeks 4, 8, and 12; a DXA scan and OGTT were repeated at week 12.

Intervention

After baseline participants were randomly assigned to a sweetener treatment (assignments generated from GraphPad Software: www.graphpad.com). Owing to the high attrition rate in the sucrose treatment group, randomization was regenerated after 100 participants to assign 2 participants to the sucrose treatment for every 1 assigned to the LCS treatments. Participants consumed between 1.25 and 1.75 L of a colored, fruit-flavored beverage (Kool-Aid, Kraft Foods) per day. Volume consumed was dependent on body weight at baseline: participants with body weight of 60–75 kg consumed 1.25 L/d, 76–90 kg consumed 1.50 L/d, and >91 kg consumed 1.75 L/d. Participants were blinded to the sweetener treatment and were provided no additional dietary recommendations besides required consumption of the intervention beverage. Participants choose their beverage flavor (lemon, orange, or mixed berry) on a weekly basis.

Sucrose beverages contained 100, 120, or 140 g of sucrose in 1.25, 1.50, or 1.75 L of water (400, 480, or 560 kcal), respectively. To put this into perspective of typical beverage consumption, mean water from beverages other than plain water is ∼1.5 L/d (32). Mean energy from sugar-sweetened beverages (SSBs) ranges from 236 to 338 kcal/d among US adults, with ≤20% of young adults consuming >500 kcal/d from SSBs (33).

LCS beverages contained 0.73, 0.58, 0.66, or 0.16 g of saccharin, aspartame, rebA, and sucralose, respectively. Sweetener concentrations of the beverages were designed to fall within acceptable daily intake (ADI) ranges set by the Food and Drug Administration and Joint FAO/WHO Expert Committee on Food Additives (34) and to match the sweetness intensity of an 8% wt:vol sucrose beverage based on predicted iso-sweet concentrations from results of a pretrial sensory test. Because sweetness potency varies by sweetener and concentration, LCS concentrations were not the same across treatments. In the sensory test, 135 participants sampled 5 quarter-log dilutions of sucrose (2.5–25%) and 5 quarter-log dilutions of 1 of 4 LCSs (0.014–0.14% saccharin, 0.017–0.17% aspartame, 0.016–0.16% rebA, and 0.0044–0.044% sucralose). Participants tasted and expectorated each sample and rated sweetness intensity on a visual analog scale (VAS) with endpoints of "not sweet" to "extremely sweet" via Qualtrics survey software. Linear models were used to determine the necessary concentration of each LCS to match the sweetness of 8% sucrose (the desired concentration of sucrose of our beverage formulation).

Anthropometrics

Body weight and total body water were measured using a Tanita Body Composition Analyzer (Model TBF-410GS, Tanita Inc.). Participants removed their shoes, socks, and excess clothing (coats, sweater, jewelry, etc.) before weighing. Height was measured at baseline with a Holtain stadiometer. DXA (GE/Lunar Prodigy DXA) was used to measure body composition at baseline and week 12. Specific body composition measures of interest included total fat mass, total fat-free mass (FFM), total tissue percentage fat, android fat mass, and gynoid fat mass.

Sensory and hedonic beverage assessment

Participants were told not to consume any food or drink other than water for ≥2 h before the beverage evaluation. They were asked to take 1 sip of the beverage and rate its sensory characteristics of intensity (sweet, salty, sour, bitter, off-flavor, mouth-drying, sweet aftertaste, bitter aftertaste, off-flavor aftertaste) and overall liking on a 100-point VAS with endpoints of "not at all" to "extremely" via a Qualtrics survey. Participants were also asked to report at what time they typically consumed the study beverages and if they consumed the beverages with other foods or drinks.

FMC testing

After participants finished consuming the beverage, they completed a survey on their subjective FMC and had hand steadiness (Impulse Counter, Lafayette Instrument Co.) and rotary tracking (Standard Rotary Pursuit, Model 30010A; Somatco) measured.

OGTT

A 2-h, 7-sample OGTT was administered at baseline and week 12 (35). Upon arrival after an overnight fast, an indwelling catheter was placed (time = −10 min). A 2-mL whole blood sample with EDTA and a 4-mL sample for serum were obtained in vacutainer tubes at time 0 immediately before consumption of the 75-g glucose drink (Azer Scientific). Subsequent blood samples were collected at minutes 10, 20, 30, 60, 90, and 120. Samples were centrifuged and the serum divided into aliquots and frozen at −80°C for batch analysis. A single fasting blood draw was collected at week 6. Whole blood glycated hemoglobin (HbA1c) and serum glucose, insulin, triacylglycerol (TG), HDL, and LDL cholesterol were determined on a Roche COBAS 400 Plus analyzer.

Dietary intake

Food and energy intake were measured on 3 d (2 nonconsecutive weekdays and 1 weekend day) during baseline and weeks 4, 8, and 12 using the web-based NIH "Automated Self-Administered 24-hour Dietary Recall" (ASA24), version 2014 system, developed by the National Cancer Institute (Bethesda, MD). If the participant failed to complete a recall, the testing day was rescheduled within 1 wk of the missed recall. The Goldberg formula was used to determine whether recorded dietary intake was physiologically plausible (36).

EE and physical activity

Free-living EE was measured using RT6 triaxial accelerometers (Higi LLC). Measurements were taken at baseline and weeks 4, 8, and 12 on the same 3 testing days that dietary intake was recorded. EE was captured every 60 s and calculated using the Katch-McArdle equation (37). Habitual work, sport, and leisure physical activity were assessed using the Baecke Questionnaire (38).

Appetitive sensations

Appetite logs were completed using a smartphone and/or web-based Qualtrics survey on one of the days participants completed a dietary recall every 4 wk. The appetite log asked participants to rate their hunger, fullness, desire to eat (DTE), prospective consumption, preoccupation with food, and thirst on a 100-point VAS (39) on the hour, every hour while awake for a 24-h period.

Compliance

Eighty milligrams of PABA were incorporated into the beverage powders daily to serve as a compliance biomarker. Not all LCSs used in this study are excreted in the urine (22); thus, urinary concentration of LCSs could not be measured directly (40). Twenty-four-hour urine samples were collected in 3-L opaque collection bottles at baseline as well as weeks 4, 8, and 12. Urinary PABA recovery was analyzed using spectrophotometric methods outlined by Sharma et al. (41). Completeness of PABA collection was calculated as the measure of compliance ($\frac{(Measured\ PABA - Baseline\ PABA)}{Observed:\ Predicted\ Creatinine}$).

Beverage Consumption Habits Questionnaire

A modified version of the Brief Questionnaire to Assess Habitual Beverage Intake (42) (referred to as the Beverage Consumption Habits Questionnaire or mBEV) measured habitual beverage intake over the past month. The mBEV questionnaire was modified to include questions assessing consumption of flavored water; vegetable juice; diet sweet tea; diet energy drinks; diet sports drinks; subdivision of "Soft Drinks, Regular" and "Diet Soft Drinks/Artificially Sweetened Drinks" into "Regular Soda, soft drink," "Diet Soda, soft drinks," "Sweetened Juice beverage/drink," and "Diet sweetened drinks"; and subdivision of "tea or coffee, black, with/without artificial sweetener" into "tea or coffee with cream or non-dairy creamer," "tea or coffee, black with LCS," and "tea or coffee, with cream or non-dairy creamer with LCS." mBEV was completed at baseline, and weeks 4, 8, and 12 to track if and how participants responded to the intervention by adding or substituting the intervention beverage into their diet. An estimate of daily energy provided from discretionary beverages (not beverages required for consumption) was calculated based on ingestion frequency, typical volume of beverage, and average energy content of the beverage (determined based on nutrient content obtained from the USDA Food Composition Database standard reference).

Week 12 questionnaire

At the end of the 12-wk trial, participants completed an additional questionnaire. Questions included what type of sweetener they thought they consumed (sugar, LCS, no sweetener, both sugar and LCS, alternating LCS and sugar) and if the study altered their desire to eat sweet foods and beverages. If they indicated that they had a change in desire for sweet foods or beverages, they indicated how strong the change in their desire for sweet was on a 100-point VAS with equi-spaced descriptors of "desire decreased extremely," "desire decreased slightly," "no change in desire," "desire increased slightly," and "desire increased extremely."

Statistical analysis

Power calculations were based on a significant change in body weight from a report by Raben et al. (43), that compared 2 groups [1 LCS ($n = 21$) and 1 sucrose ($n = 20$)] who adhered to a LCS intervention for 10 wk. The present experiment was 2 wk longer and sampled >25 individuals per treatment group. We considered a variance slightly larger (~4%) than that observed (SD = 1.78 kg) in the Raben et al. report (43) to reflect its expected increase given the additional 2 wk of intervention and used a significance level of $\alpha < 0.05$. With this variance and stated sample size, we had 80% power to detect differences in average weight gain of 2.8 kg.

One-factor ANOVA was used to compare baseline characteristics between treatment groups; Pearson's chi-squared test was used to compare gender distributions and attrition between treatments. Linear mixed models were used to determine the effect of time, treatment, and the treatment-by-time interaction on change in body weight, absolute body weight, and BMI. Additional linear mixed models controlling for baseline body weight were used to determine the effect of time, treatment, and time-by-treatment interaction on body weight and body composition. Linear mixed models were used to determine the effect of time, treatment, and treatment-by-time interaction on dietary intake, EE, appetitive sensations, beverage sensory attributes, and hedonics as well. Sweetener treatment and time were fixed effects. Participants were treated as random effects repeated over time using a Toeplitz repeated covariance matrix. Statistical significance was determined by $\alpha < 0.05$. Least significant difference (LSD) was used for post hoc comparisons.

The median PABA recovery from each participant during weeks 4, 8, and 12 was used to assess compliance. A Kruskal–Wallis H test was used to determine if median urinary PABA recovery was different between treatments. Statistical significance was determined by $\alpha < 0.05$.

One outlier (defined as outside the 25th and 75th percentile $\pm$ 3.1 $\times$ IQR) in the aspartame treatment group was

TABLE 1 Baseline characteristics[1]

	Sucrose	Saccharin	Aspartame	RebA	Sucralose	*P* value
n	39	29	30	28	28	
Age, y	28.2 ± 9.5	25.8 ± 6.9	29.5 ± 12.0	27.1 ± 9.6	25.9 ± 9.0	0.538
Weight, kg	90.2 ± 18.1	84.4 ± 16.9	87.6 ± 15.7	84.4 ± 12.3	82.9 ± 10.6	0.278
Height, cm	171.6 ± 10.2	170.8 ± 9.9	170.2 ± 10.8	168.1 ± 9.1	170.2 ± 10.1	0.720
BMI, kg/m^2	30.4 ± 4.1	28.8 ± 4.4	30.3 ± 5.3	29.9 ± 3.8	28.7 ± 4.0	0.368
Female, %	53.8	58.6	50.0	64.3	57.1	0.848
Dropouts (count)	16	4	4	3	4	0.006*
TFEQ scores						
Restraint	6.4 ± 3.3	7.8 ± 3.7	6.9 ± 3.2	6.8 ± 4.1	6.7 ± 3.6	0.661
Disinhibition	6.8 ± 3.8	5.5 ± 3.3	6.1 ± 2.7	6.8 ± 3.6	6.4 ± 3.2	0.565
Hunger	5.5 ± 3.3	4.6 ± 3.5	4.5 ± 3.1	5.3 ± 3.9	4.3 ± 2.9	0.564
Baecke						
Work index	2.3 ± 0.4	2.4 ± 0.5	2.2 ± 0.6	2.4 ± 0.6	2.1 ± 0.6	0.136
Sport index	2.4 ± 0.7	2.5 ± 0.7	2.5 ± 1.1	2.5 ± 0.8	2.7 ± 0.9	0.545
Leisure index	2.8 ± 0.6	2.8 ± 0.5	2.6 ± 0.6	2.7 ± 0.5	2.7 ± 0.8	0.645

[1] Values are means ± SDs. One-factor ANOVA was used to compare baseline characteristics between treatment groups; Pearson's chi-squared test was used to compare gender distributions and attrition between treatments. *Significant difference between treatments ($P < 0.05$). Baecke, Baecke Physical Activity Questionnaire (which includes indexes of work, sport, and leisure physical activity); RebA, rebaudioside A; TFEQ, Three Factor Eating Questionnaire (which includes scores for dietary restraint, disinhibition, and hunger).

removed before analysis of EE. Values for the appetite logs were reported as mean appetitive sensation from the 24-h appetite logs. Logs with <4 entries were excluded from analysis (data from 4 participants were excluded). Baseline urinary PABA data were missing for 1 participant owing to a missed collection. Mean PABA at baseline of the total sample was used to replace the missing value.

Only participants who completed the OGTT at week 0 and week 12 were included in the statistical analysis of blood metabolites ($n = 123$). Repeated-measures ANOVA was used to compare blood glucose and insulin concentrations from the OGTTs and values of incremental AUC (iAUC), HbA1c, total cholesterol, HDL, LDL, and serum TG. Participants were treated as random effects. For blood glucose and insulin, the main effects were time point during the 2-h test (time), sweetener treatment (treatment), session (baseline and week 12), treatment-by-time, treatment-by-session, and treatment-by-session-by-time. When sample values were missing, the value was replaced with the mean value for all treatment groups at the missing time point. Significant main effects of treatment, time, and the interaction of treatment-by-time were tested for HbA1c, total cholesterol, HDL, LDL, and serum TG.

Data are expressed as mean ± SE unless otherwise stated. IBM SPSS Statistics version 24 was used to complete the statistical analysis with the exception of the blood glucose or insulin iAUC calculation where SAS software, version 9.4 (SAS Institute Inc., Cary, NC) was used.

Results

Baseline characteristics

Baseline characteristics for the 154 participants who enrolled in the trial are shown in **Table 1**. There were no significant group differences for baseline anthropometric measures, age, gender, Three Factor Eating Questionnaire subscores, and Baecke Physical Activity Questionnaire indexes ($P \geq 0.14$). There were significantly more participants who discontinued participation in the study from the sucrose group than from the other sweetener groups ($P = 0.006$).

Compliance

Median urinary PABA recovery corrected for urinary creatinine was 63.3%, 71.5%, 71.5%, 70.0%, and 82.6% for the sucrose, saccharin, aspartame, rebA, and sucralose treatment groups, respectively. PABA recovery was not significantly different between treatments ($P = 0.37$). Follow-up experiments indicated that incomplete PABA recovery was likely due to a lack of preservative [such as boric acid (41)] during urine collection.

Sensory properties of the beverages and overall liking

The beverages used in this trial were not significantly different in perceived intensity of sweet, salty, sour, bitter, off-flavor, drying, bitter aftertaste, and off-flavor aftertaste between treatments ($P \geq 0.10$) or between treatments over time ($P \geq 0.10$). Overall liking was not significantly different between beverage treatments or between treatments over time ($P \geq 0.52$).

Body weight change

Change in body weight was significantly different among the 5 sweetener groups (treatment-by-time effect, $P = 0.01$). There was a statistically significant increase in body weight over the 12-wk trial for participants in the sucrose and saccharin treatments (weight change = 1.85 ± 0.36 kg, $P < 0.001$; 1.18 ± 0.36 kg, $P = 0.02$, respectively) (**Figure 1**). The nonsignificant changes in body weight for aspartame, rebA, and sucralose were 0.73 ± 0.35 kg, 0.60 ± 0.36 kg, and −0.78 ± 0.36 kg, respectively ($P \geq 0.07$). Body weight at baseline among the sucrose group tended to be greater than the sucralose treatment ($P = 0.05$). An analysis of BMI and body weight controlling

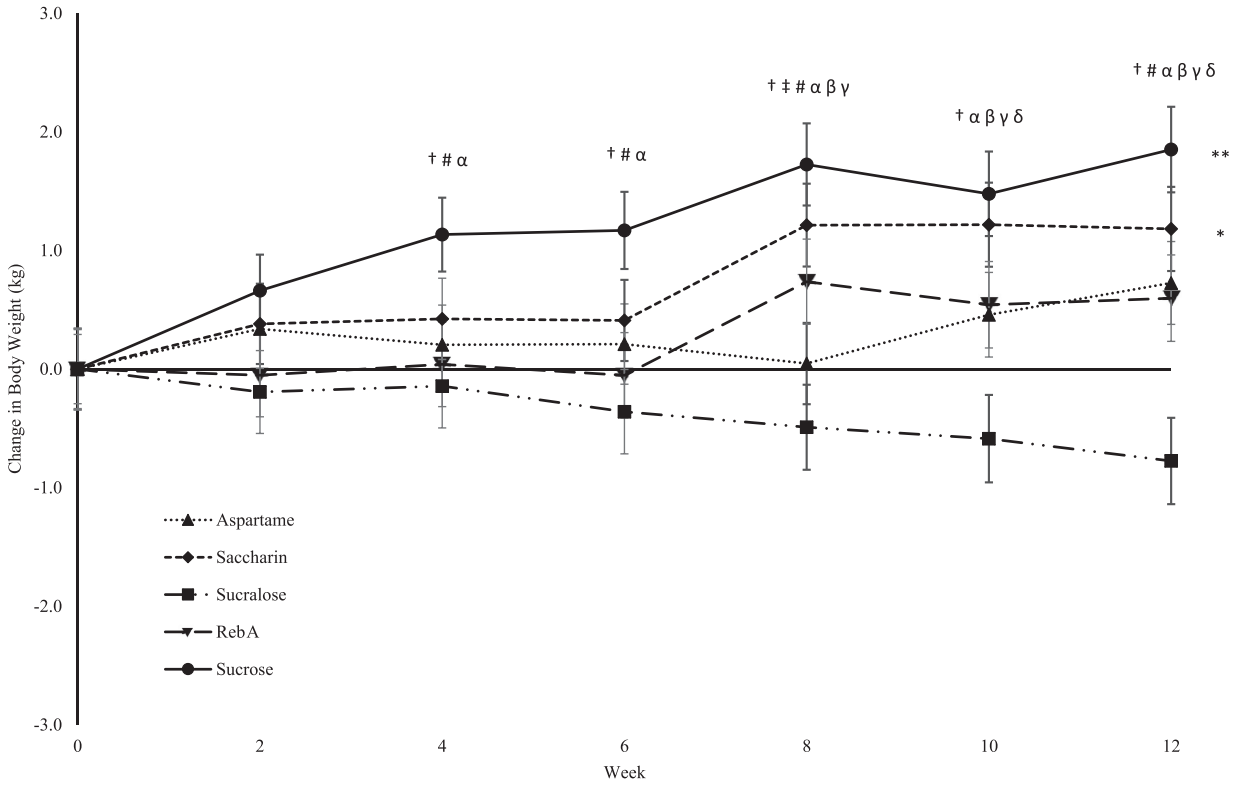


FIGURE 1 Change in body weight with sweetener consumption for 12 wk. Data presented as mean ± SE. $n = 39, 29, 30, 28, 28$ for sucrose, saccharin, aspartame, rebA, and sucralose groups, respectively. A linear mixed model was used to determine the main effects of treatment and time and the interaction of treatment-by-time on change in body weight. Treatment, time, and treatment-by-time interaction were statistically significant ($P \leq 0.01$). Change in body weight over time was statistically significant for the sucrose ($^{**}P < 0.001$) and saccharin ($^{*}P = 0.02$) treatments. The following symbols above the data points denote significant differences between treatments within a week: †, aspartame significantly different than sucrose ($P < 0.05$); ‡, aspartame significantly different than saccharin ($P < 0.05$); #, rebA sucralose significantly different than sucrose ($P < 0.05$); α, sucralose significantly different than sucrose ($P < 0.05$); β, sucralose significantly different than saccharin ($P < 0.05$); γ, sucralose significantly different than rebA; δ, sucralose significantly different than aspartame. RebA, rebaudioside A.

for baseline body weight followed a similar pattern of increased BMI and body weight with sucrose and saccharin consumption ($P \leq 0.04$) and no significant change with aspartame, rebA, and sucralose consumption ($P \geq 0.07$) (**Table 2**). Therefore, changes in body weight were not simply an artifact of individuals with higher initial weight gaining more weight. Mean change in body weight was not significantly different between sucrose and saccharin ($P = 0.21$) and both were significantly greater than sucralose (+1.51 kg, $P < 0.001$ for sucrose and +1.05 kg, $P = 0.01$ for saccharin). Mean body weight change was also significantly higher for sucrose than for aspartame and rebA ($P \leq 0.02$). Mean change in body weight was not significantly different between aspartame, rebA, and sucralose ($P \geq 0.09$). Between the LCS treatments, change in body weight at week 12 was statistically lower among the sucralose treatment than for the saccharin (difference = −1.96 kg, $P < 0.001$), aspartame (−1.50 kg, $P = 0.003$), and rebA (−1.37 kg, $P = 0.008$) groups.

Body compositions

Total fat mass (not controlling for baseline body weight) increased significantly during the intervention among participants in the sucrose treatment ($\Delta = +1.35 \pm 0.25$ kg, $P < 0.001$). At week 12, change in total fat mass was significantly higher for the sucrose group than for all LCSs ($P \leq 0.01$). Among the LCS groups, change in total fat mass was negative and significantly lower for the sucralose group than for the saccharin and aspartame groups (−0.79 ± 0.33 kg and −0.80 ± 0.33 kg, respectively; $P \leq 0.02$). Total fat mass controlling for baseline body weight (Table 2) was significantly higher for the sucrose group than for the sucralose group ($P = 0.045$) with no difference in total fat mass between the other sweetener groups.

Total FFM significantly increased among the sucrose, saccharin, and aspartame groups (+0.84 ± 0.20 kg, $P = 0.001$ for sucrose; +0.70 ± 0.18 kg, $P = 0.006$ for saccharin; +0.63 ± 0.18 kg, $P = 0.01$ for aspartame). At week 12, change in FFM was significantly greater among the sucrose group than among the rebA and sucralose groups (+0.60 ± 0.27 kg, $P = 0.03$ compared with rebA; +1.17 ± 0.27 kg, $P < 0.001$ compared with sucralose). Change in FFM among the sucralose group was negative and significantly lower than other LCS groups ($P \leq 0.03$). Controlling for baseline body weight, total FFM increased significantly among the sucrose, saccharin, and aspartame groups (+0.85 ± 0.29 kg, $P = 0.004$ for sucrose; +0.71 ± 0.28 kg, $P = 0.01$ for saccharin; +0.63 kg ± 0.27, $P = 0.02$ for aspartame).

TABLE 2 Body composition at baseline and week 12[1]

							P values		
		Sucrose	Saccharin	Aspartame	RebA	Sucralose	Treatment	Time	Treatment × time
Body weight, kg	Week 0	90.24 ± 2.46	84.38 ± 2.85	87.57 ± 2.81	84.36 ± 2.90	82.86 ± 2.90	0.157	0.005	0.012
	Week 12	92.04 ± 2.48*[1]	85.55 ± 2.86*[1,2]	88.23 ± 2.81[1,2]	84.92 ± 2.91[1,2]	82.10 ± 2.91[2]			
Δ Body weight, kg	—	1.85 ± 0.36*[1]	1.18 ± 0.36*[1,2]	0.73 ± 0.35[2]	0.60 ± 0.36[2]	−0.78 ± 0.36[3]	0.002	0.003	0.010
Body weight, kg, adjusted†	Week 0	86.24 ± 0.29	86.24 ± 0.34	86.24 ± 0.33	86.24 ± 0.35	86.25 ± 0.35	0.002	0.003	0.010
	Week 12	88.09 ± 0.36*[1]	87.43 ± 0.36*[1,2]	86.97 ± 0.35[2]	86.84 ± 0.37[2]	85.47 ± 0.37[3]			
BMI, kg/m^2	Week 0	30.44 ± 0.70	28.83 ± 0.81	30.32 ± 0.80	29.86 ± 0.83	28.72 ± 0.83	0.237	0.006	0.021
	Week 12	31.04 ± 0.71*[1]	29.23 ± 0.81*[1,2]	30.52 ± 0.80[1,2]	30.04 ± 0.83[1,2]	28.46 ± 0.83[2]			
Δ BMI, kg/m^2	—	0.62 ± 0.12*[1]	0.41 ± 0.12*[1,2]	0.22 ± 0.12[2]	0.20 ± 0.12[2]	−0.27 ± 0.12[3]	0.002	0.003	0.015
BMI, kg/m^2, adjusted†	Week 0	29.60 ± 0.48	29.22 ± 0.55	30.04 ± 0.54	30.25 ± 0.56	29.44 ± 0.56	0.653	0.007	0.022
	Week 12	30.18 ± 0.49*	29.63 ± 0.55*	30.24 ± 0.54	30.44 ± 0.56	29.18 ± 0.57			
Total fat mass, kg	Week 0	34.35 ± 1.79[1]	30.70 ± 2.07[1,2]	32.24 ± 2.04[1,2]	32.10 ± 2.11[1,2]	28.10 ± 2.11[2]	0.149	0.001	0.032
	Week 12	35.68 ± 1.81*[1]	31.16 ± 2.08[1,2]	32.72 ± 2.04[1,2]	32.19 ± 2.11[1,2]	27.80 ± 2.11[2]			
Δ Total fat mass, kg	—	1.35 ± 0.25*[1]	0.48 ± 0.23[2]	0.49 ± 0.23[2]	0.09 ± 0.23[2,3]	−0.31 ± 0.24[3]	0.006	0.004	0.006
Total fat mass, kg, adjusted†	Week 0	32.26 ± 1.29	31.62 ± 1.49	31.52 ± 1.46	33.03 ± 1.52	29.80 ± 1.52	0.488	0.012	0.033
	Week 12	33.58 ± 1.32* [1]	32.08 ± 1.49[1,2]	32.01 ± 1.47[1,2]	33.11 ± 1.52[1,2]	29.49 ± 1.53[2]			
Total fat-free mass, kg	Week 0	55.62 ± 1.73	53.72 ± 2.01	55.04 ± 1.97	52.08 ± 2.04	54.21 ± 2.04	0.673	0.001	0.030
	Week 12	56.47 ± 1.74*	54.43 ± 2.01*	55.67 ± 1.97*	52.33 ± 2.04	53.89 ± 2.04			
Δ Total fat-free mass, kg	—	0.84 ± 0.20*[1]	0.70 ± 0.18*[1,2]	0.63 ± 0.18*[1,2]	0.25 ± 0.18[2]	−0.33 ± 0.19[3]	0.009	<0.001	0.009
Total fat-free mass, kg, adjusted†	Week 0	53.68 ± 1.30	54.58 ± 1.49	54.38 ± 1.47	52.95 ± 1.52	55.80 ± 1.52	0.802	0.001	0.028
	Week 12	54.54 ± 1.31*	55.29 ± 1.50*	55.01 ± 1.47*	53.20 ± 1.52	55.46 ± 1.53			
Total tissue % fat	Week 0	39.08 ± 1.55	37.22 ± 1.80	37.57 ± 1.77	39.11 ± 1.83	35.18 ± 1.83	0.434	0.558	0.476
	Week 12	39.65 ± 1.57	37.20 ± 1.80	37.62 ± 1.77	39.07 ± 1.83	35.01 ± 1.84			
Δ Tissue % fat	—	0.57 ± 0.20	−0.01 ± 0.19	0.06 ± 0.19	−0.04 ± 0.19	−0.17 ± 0.19	0.289	0.482	0.289
Total tissue % fat, adjusted†	Week 0	38.54 ± 1.54	37.46 ± 1.77	37.38 ± 1.74	39.36 ± 1.80	35.63 ± 1.81	0.587	0.562	0.478
	Week 12	39.10 ± 1.55	37.44 ± 1.77	37.44 ± 1.74	39.31 ± 1.80	35.46 ± 1.81			
Android fat mass, kg	Week 0	3.22 ± 0.22	2.69 ± 0.26	3.01 ± 0.26	2.91 ± 0.26	2.33 ± 0.26	0.076	0.075	0.054
	Week 12	3.38 ± 0.23	2.75 ± 0.26	3.03 ± 0.26	2.89 ± 0.27	2.31 ± 0.27			
Δ Android fat mass, kg	—	0.16 ± 0.03*[1]	0.06 ± 0.03[2]	0.02 ± 0.03[2]	−0.03 ± 0.03[2]	−0.02 ± 0.03[2]	0.012	0.047	0.012
Android fat mass, kg, adjusted†	Week 0	2.94 ± 0.15	2.82 ± 0.17	2.92 ± 0.17	3.04 ± 0.18	2.56 ± 0.18	0.278	0.080	0.058
	Week 12	3.10 ± 0.15	2.87 ± 0.17	2.94 ± 0.17	3.01 ± 0.18	2.54 ± 0.18			
Gynoid fat mass, kg	Week 0	5.75 ± 0.32	5.43 ± 0.37	5.38 ± 0.36	5.66 ± 0.38	4.96 ± 0.38	0.411	0.044	0.004
	Week 12	6.03 ± 0.32*[1]	5.54 ± 0.37[1,2]	5.41 ± 0.36[1,2]	5.60 ± 0.38[1,2]	4.90 ± 0.38[2]			
Δ Gynoid fat mass, kg	—	0.28 ± 0.05*[1]	0.12 ± 0.04[2]	0.04 ± 0.04[2,3]	−0.06 ± 0.04[3]	−0.06 ± 0.05[3]	<0.001	0.021	<0.001
Gynoid fat mass, kg, adjusted†	Week 0	5.45 ± 0.27	5.56 ± 0.31	5.27 ± 0.30	5.79 ± 0.31	5.21 ± 0.31	0.637	0.048	0.004
	Week 12	5.73 ± 0.27*	5.67 ± 0.31	5.31 ± 0.30	5.73 ± 0.31	5.14 ± 0.31			
TBW, kg	Week 0	44.39 ± 1.42	42.05 ± 1.58	43.66 ± 1.56	40.98 ± 1.64	42.18 ± 1.67	0.542	0.132	0.863
	Week 12	44.31 ± 1.43	42.24 ± 1.59	43.08 ± 1.56	40.77 ± 1.64	41.76 ± 1.67			
Δ TBW, kg	—	−0.07 ± 0.35	0.18 ± 0.36	−0.60 ± 0.36	−0.22 ± 0.37	−0.41 ± 0.37	0.167	0.188	0.912
TBW, kg, adjusted†	Week 0	42.61 ± 1.06	42.79 ± 1.16	43.19 ± 1.14	41.55 ± 1.21	43.35 ± 1.23	0.884	0.131	0.859
	Week 12	42.54 ± 1.07	42.98 ± 1.17	42.62 ± 1.15	41.34 ± 1.21	42.93 ± 1.23			

[1]Values are mean ± SE. $n = 39, 29, 30, 28, 28$ for sucrose, saccharin, aspartame, rebA, and sucralose groups, respectively. Linear mixed model of main effects of treatment, time, and the interaction of treatment-by-time on anthropometrics and change in anthropometrics. †Linear mixed models of main effects of treatment, time, and the interaction of treatment-by-time on anthropometrics controlling for body weight at baseline. Treatments within a week without a common number differ, $P < 0.05$. *Significant difference ($P < 0.05$) across weeks within each treatment. RebA, rebaudioside A; TBW, total body water.

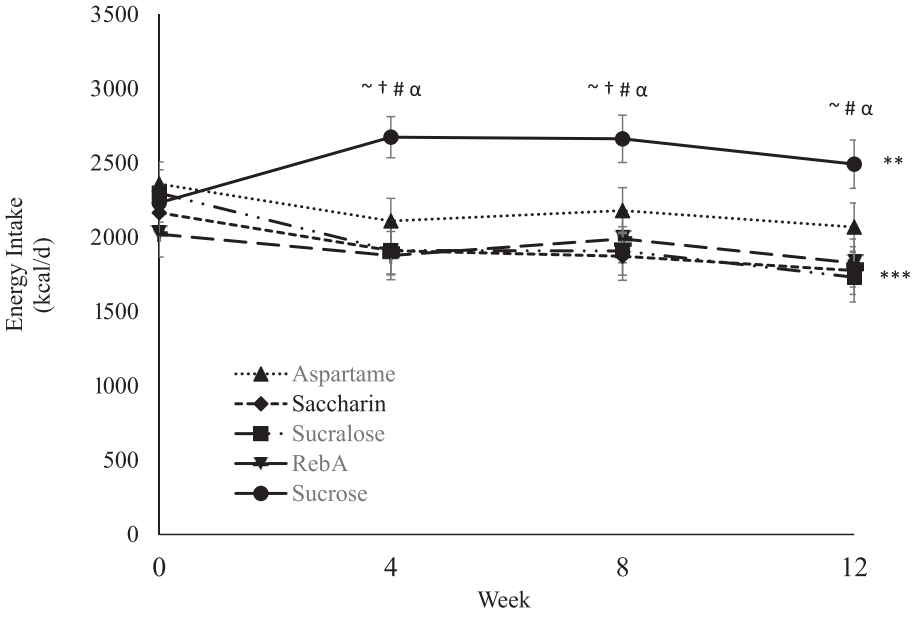


FIGURE 2 Change in energy intake with sweetener consumption for 12 wk. Data presented as mean ± SE. $n = 39, 29, 30, 28, 27$ for sucrose, saccharin, aspartame, rebA, and sucralose groups, respectively. A linear mixed model was used to determine the main effects of treatment and time and the interaction of treatment-by-time on change in energy intake. Treatment, time, and treatment-by-time were statistically significant ($P \leq 0.04$). Change in energy intake over time was statistically significant for the sucrose (**$P = 0.007$) and sucralose (***$P = 0.02$) treatments. The following symbols above the data points denote significant differences between treatments within a week: ~, saccharin significantly different than sucrose ($P < 0.01$); †, aspartame significantly different than sucrose ($P < 0.05$); #, rebA sucralose significantly different than sucrose ($P < 0.01$); α, sucralose significantly different than sucrose ($P < 0.01$). RebA, rebaudioside A.

Android and gynoid fat mass increased significantly during the intervention among participants in the sucrose treatment (+0.16 ± 0.03 kg and +0.28 ± 0.05 kg, respectively; $P < 0.001$). At week 12, changes in android fat mass and gynoid fat mass in the sucrose group were significantly greater than in all LCS groups ($P \leq 0.03$, $P \leq 0.01$, respectively). There were no significant differences in change in android fat mass between the LCS groups. Change in gynoid fat mass was significantly higher among the saccharin group than among the rebA and sucralose treatment groups [+0.18 ± 0.06 kg for both rebA ($P = 0.006$) and sucralose ($P = 0.005$)]. After controlling for baseline body weight, gynoid fat mass significantly increased in the sucrose group ($P < 0.001$) with no significant difference among any of the LCS treatment groups ($P \geq 0.18$). The treatment, time, and treatment-by-time effects were nonsignificant for android fat mass ($P \geq 0.06$).

Total body water was not significantly different between treatments, over time, or between treatments over time ($P \geq 0.13$) with and without controlling for baseline body weight.

Energy intake

Self-reported energy intake data from the ASA24 are shown in **Figure 2**. Energy intake at baseline was not significantly different between treatments ($P \geq 0.11$). Reported energy intake increased with sucrose consumption and decreased with sucralose consumption throughout the 12-wk trial ($P = 0.007$ and 0.02, respectively). Mean reported energy intake for the sucrose treatment group was significantly higher than for the saccharin, aspartame, rebA, and sucralose treatment groups by 584 ± 162, 336 ± 160, 587 ± 164, and 553 ± 164 kcal, respectively ($P \leq 0.04$). At week 12, energy intake was significantly higher for the sucrose group than for the saccharin, rebA, and sucralose treatment groups ($P \leq 0.004$) and only tended to be greater than the aspartame treatment group ($P = 0.07$). The difference in energy intake between the sucrose and LCS groups is attributed to the energy provided from the beverage. When the energy provided by the beverage is removed from the reported energy intake, the treatment and treatment-by-time main and interaction effects are not statistically significant ($P = 0.37$ and 0.96, respectively). There was no significant difference in energy intake between the LCS groups at any time point. Notably, only 14.2% of the mean reported energy intakes fell within the Goldberg cutoffs.

mBEV results

Mean energy provided from discretionary beverages as estimated from the mBEV was 249 ± 20.7, 205 ± 21.1, 183 ± 22.3, and 208 ± 21.6 kcal at week 0, 4, 8, and 12, respectively ($P = 0.003$). Treatment and time-by-treatment effects were not statistically significant ($P = 0.64$ and 0.76, respectively).

Ingestive frequency

Ingestive frequency and portion size data from the ASA24 are displayed in **Table 3**. Self-reported number of eating events ≥100 kcal was used to determine ingestive frequency. Ingestive frequency increased significantly in the sucrose group and significantly decreased in the sucralose group ($P < 0.001$ and $P = 0.005$, respectively). Ingestive frequency was significantly

TABLE 3 Ingestive frequency and portion size with sweetener consumption for 12 wk[1]

							P values		
		Sucrose	Saccharin	Aspartame	RebA	Sucralose	Treatment	Time	Treatment × time
Ingestive frequency (occasions/d)	Week 0	3.4 ± 0.2[1]	3.7 ± 0.2[1,2]	4.0 ± 0.2[2]	3.4 ± 0.2[1,2]	3.6 ± 0.2[1,2]	<0.001	0.993	<0.001
	Week 4	4.8 ± 0.2*[1]	3.5 ± 0.2[2,3]	3.6 ± 0.2[2]	3.1 ± 0.2[2,3]	3.0 ± 0.2*[3]			
	Week 8	4.9 ± 0.2*[1]	3.4 ± 0.2[2]	3.5 ± 0.2*[2]	3.3 ± 0.2[2]	2.9 ± 0.2*[2]			
	Week 12	4.8 ± 0.2*[1]	3.6 ± 0.2[2]	3.5 ± 0.2[2,3]	3.2 ± 0.2[2,3]	2.9 ± 0.2*[3]			
Portion size per ingestive event with beverage (kcal/event)	Week 0[a]	676.6 ± 35.2	596.5 ± 40.8	613.5 ± 40.1	620.5 ± 41.5	633.9 ± 42.3	0.093	0.021	0.138
	Week 4[a,b]	566.3 ± 37.2	521.8 ± 42.6	627.8 ± 41.3	651.3 ± 44.2	659.2 ± 42.9			
	Week 8[a]	550.0 ± 43.5	533.4 ± 44.0	688.6 ± 41.9	627.5 ± 43.5	687.2 ± 44.2			
	Week 12[b]	519.1 ± 43.5	505.8 ± 43.3	614.9 ± 43.0	586.8 ± 43.5	605.3 ± 44.9			
Portion size per eating event without beverage (kcal/event)	Week 0[a,b]	676.6 ± 36.4	596.5 ± 42.2	613.5 ± 41.5	620.5 ± 43.0	633.9 ± 43.8	0.002	0.048	0.577
	Week 4[a,b]	738.6 ± 38.5	522.1 ± 44.1	627.8 ± 42.7	651.6 ± 45.7	659.2 ± 44.4			
	Week 8[a]	782.0 ± 44.9	532.9 ± 45.4	688.2 ± 43.3	627.3 ± 45.0	686.7 ± 45.8			
	Week 12[b]	699.5 ± 44.7	505.9 ± 44.7	615.1 ± 44.4	586.5 ± 44.9	604.1 ± 46.4			

[1]Values are mean ± SE. $n = 39, 29, 30, 28, 27$ for sucrose, saccharin, aspartame, rebA, and sucralose groups, respectively. Linear mixed model of main effects of treatment and time and the interaction of treatment-by-time on change in ingestive frequency and portion size. Ingestive frequency: main effect of treatment statistically significant. The ingestive frequency of sucrose treatment was greater than all low-calorie sweetener groups ($P < 0.01$); aspartame group was significantly higher than sucralose ($P = 0.03$). Portion size without beverage: main effect of treatment statistically significant. Portion size without beverage of sucrose group significantly greater than saccharin, aspartame, and rebA groups ($P < 0.05$); aspartame significantly higher than saccharin ($P = 0.04$); rebA significantly higher than saccharin ($P = 0.02$); sucralose significantly higher than saccharin ($P = 0.03$). Weeks without a common letter differ, $P < 0.05$. Treatments within a week without a common number differ, $P < 0.05$. *Significant difference ($P < 0.05$) compared with values at Week 0 within each treatment. RebA, rebaudioside A.

greater among the sucrose group than among the LCS groups at weeks 4, 8, and 12 ($P \leq 0.001$). Ingestive frequency was higher for the aspartame group than for sucrose at baseline ($P = 0.03$). Ingestive frequency among the sucralose group was significantly lower than saccharin at week 12 (difference = −0.70 events/d, $P = 0.04$), but did not differ from the other groups at any other weeks.

Portion size

When energy provided from the sucrose beverages was included, mean portion size was 628 ± 17.9, 605 ± 18.7, 617 ± 19.4, and 566 ± 19.5 kcal at weeks 0, 4, 8, and 12, respectively ($P = 0.02$). Treatment and treatment-by-time effects were not statistically significant ($P = 0.09$ and 0.14, respectively). Analyses of portion size determined by energy provided from foods and beverages other than the study beverages revealed the sucrose treatment was associated with significantly greater portions than the saccharin, aspartame, and rebA treatments ($P < 0.05$) and tended to be greater than the sucralose treatment ($P = 0.09$). Participants who consumed saccharin reported significantly lower portion sizes than the aspartame and sucralose groups ($P \leq 0.04$) and tended to be lower than the rebA treatment ($P = 0.08$). The treatment-by-time interaction effect was not statistically significant ($P = 0.58$).

Glycemic response

Results from the 123 participants who completed the OGTT at baseline and week 12 are presented in **Supplemental Figure 2**. There was no significant treatment effect ($P = 0.08$), although mean glucose concentrations during the OGTT increased from baseline levels to week 12 ($P = 0.02$). Treatment-by-time, treatment-by-session, session-by-time, and treatment-by-session-by-time interaction effects were not statistically significant ($P > 0.34$). The main effects of sweetener, time, and the interaction of sweetener-by-time on glucose iAUC were not statistically significant ($P = 0.06$, 0.18, and 0.74, respectively). Insulin concentrations did not differ between baseline and week 12 during the OGTT, and there was no significant treatment effect ($P = 0.85$ and 0.27, respectively). The main effects of sweetener, time, and the interaction of sweetener-by-time on insulin iAUC were not statistically significant ($P = 0.62$, 0.28, and 0.10, respectively). Fasting serum glucose and insulin were not different between treatment groups at baseline, week 6, or week 12 ($P \geq 0.21$ and $P \geq 0.47$, respectively).

Total cholesterol and LDL cholesterol at week 12 significantly decreased from baseline ($P < 0.05$) with no difference between treatments or between treatments over time ($P \geq 0.54$). HbA1c, HDL, and serum TG were not affected by the intervention.

EE and physical activity

There was no significant treatment effect or treatment-by-time interaction effect on EE ($P = 0.56$ and 0.17, respectively). There was no difference in the Baecke indexes for work, sport, or leisure physical activity between treatments, over time, or between treatments over time ($P \geq 0.05$).

Appetite

Appetitive ratings throughout the trial are shown in **Supplemental Figure 3**. Hunger ratings among the saccharin sweetener group were significantly greater than for all other treatment groups ($P \leq 0.03$). The main effect of treatment was not significant for any other appetitive sensation ($P \geq 0.12$). DTE and prospective consumption ratings increased significantly from baseline only for the saccharin treatment group. The time-by-treatment interaction was not statistically significant for any other appetitive sensation ($P \geq 0.09$).

Week 12 questionnaire

Of participants, 57%, 69%, 52%, and 56% correctly identified whether their beverage contained a LCS in the saccharin, aspartame, rebA, and sucralose groups, respectively; 42% of participants assigned to consume beverages sweetened with sucrose correctly reported their beverage was sugar sweetened. The likelihood of correctly identifying the beverage sweetener was not different between treatment groups ($P = 0.44$). Only individuals in the aspartame treatment were able to correctly identify their beverage contained an LCS at greater than chance probability ($P < 0.05$). In the DTE sweet foods, 53% of participants reported no change, whereas 31% and 17% reported a decrease and increase, respectively; 47%, 41%, and 12% of individuals reported no change in, decreased, and increased DTE sweet beverages, respectively. Changes in the DTE sweet items were not significantly different across sweetener groups and were not associated with change in body weight ($P = 0.28$).

Discussion

Numerous RCTs have investigated the effects of individual and the combination of LCSs on body weight. The most comprehensive meta-analyses of these RCTs indicate body weight is slightly but significantly lower with LCS use (2, 3). Conversely, cohort studies reveal no association or a slight positive relation (2, 3). These weak and inconsistent findings could reflect dissimilar effects of individual LCSs, with the beneficial effects of one LCS offset by other LCSs. Studies contrasting the long-term effects of individual LCSs on body weight are lacking. This trial required healthy nondieting adults with overweight or obesity to consume a beverage containing sucrose or 1 of 4 LCSs for 12 wk. Sucrose and saccharin consumption significantly increased body weight compared with aspartame, rebA, and sucralose and weight change was directionally negative and lower for sucralose compared with saccharin, aspartame, and rebA. This differential response between LCSs indicates their consumption likely exerts effects beyond the contribution of sweetness with negligible energy. The present findings raise questions about viewing the diverse chemicals that serve as LCSs in aggregate.

Evidence for the effect of saccharin consumption on energy intake and body weight is mixed. Prospective cohort trials conclude saccharin consumption is associated with weight gain

(44, 45) although a causal association cannot be determined in these trials. In these reports, no other LCSs were evaluated exclusively because other LCSs were not commercially available or were minimally consumed at the time of data collection. Consequently, these trials do not permit determination of whether an association between saccharin intake and weight gain is unique to this sweetener. Saccharin-supplemented diets increase energy intake and body weight in select rodent studies, whereas others find no difference compared with water [reviewed in (2)]. The present findings are consistent with the positive association in epidemiological and rodent trials. Collectively, the evidence suggests saccharin stands apart from the other popular LCSs in that it may actually contribute to weight gain. A treatment-independent weight gain among the study population, estimated as 0.5–0.6 kg annually for adults (0.12–0.15 kg/3 mo) (46) and 1.5 kg annually among college students (0.38 kg/3 mo) (47), could account for a portion of the change. However, the gain noted here was 3- to 10-fold higher than this background trend. Although saccharin consumption increased total body weight in this trial, indexes of fat mass (total, android, and gynoid fat mass) were positive but not statistically significant. Effects on other health outcomes often related to adiposity, such as glucose tolerance or serum lipid concentrations, were also nonsignificant, but interpretation of this should be tempered by the short duration of this trial. Replication of the present findings and identification of the underlying mechanisms will be required to accept the hypothesis that saccharin specifically is problematic for weight management.

Sucralose consumption did not significantly alter body weight from baseline, but consumers' weights were trending downward and were significantly lower than all other treatments at week 12. Sucralose is the most commonly consumed LCS within the United States (48); yet, to our knowledge, only 1 trial has examined the effect of sucralose consumption as an exclusive LCS on body weight (49). The authors concluded that 12 wk of consumption of ≤500 mg of sucralose/d did not lead to a change of body weight. The trial was designed to measure sucralose tolerance (not body weight), the dose was ∼5 times the estimated daily intake, and was delivered as a 35-g/L solution. The extent to which sucralose consumption facilitates weight loss requires additional documentation.

The effect of aspartame consumption on body weight has been tested in multiple human feeding trials. They consistently document no effect compared with water or a no-treatment control (50–52). In the present trial, weight change with aspartame consumption was not statistically significant. The slight increase in body weight observed (+0.73 kg) may reflect the background trend in body weight but is still ∼2–5 times that rate. Unfortunately, the present trial did not include a water control to test this hypothesis.

RebA consumption had no effect on body weight. Few RCTs have investigated the effects of consumption of rebA or other steviol glycosides on health outcomes, yet the trials that did monitor body weight found no significant change (53, 54). Like aspartame, the slight increase in body weight (+0.60 kg) may be partially attributed to a rebA-independent trend for weight gain in the general population.

Sucrose consumption, particularly in the form of SSBs, is positively associated with weight gain in RCTs and epidemiological trials [reviewed in (55)]. Our trial is consistent with these findings. Using the NIH Body Weight Planner (56), the estimated increase in body weight based on baseline characteristics assuming 0% dietary compensation and 100% compliance was +5.0 kg and +4.4 kg for males and females, respectively. Accounting for the compliance from the PABA collection (63% among the sucrose treatment), the predicted change in body weight is +3.1 kg and +2.8 kg, respectively. This predicted weight change is higher than the observed change in body weight among the sucrose group (+1.85 kg). The difference may be attributable to partial dietary compensation. The partial compensation for energy derived from sugars in beverages is well documented (57, 58). This may stem from weaker satiety effects of beverage energy relative to more viscous or solid foods (59, 60). A recent meta-analysis indicates that the mandatory *addition* of SSBs to the diet is associated with increased body weight, but the recommended *reduction* of SSBs is not associated with BMI in effectiveness studies (61). Reduction of SSBs alone decreases dietary energy but also palatability. This may result in lower diet adherence and a suboptimal weight management outcome. Thus, select LCS beverages may hold advantages for energy restriction over reduction of SSBs alone by maintaining diet acceptance and improving compliance.

The present study was designed to document effects of selected sweeteners on body weight, not to identify mechanisms. Nevertheless, one notable mechanism relates to energy balance; i.e., nutritive sweeteners provide energy directly, whereas LCSs displace energy sources in the diet. There was no change in estimated energy provided from discretionary beverages throughout the trial, suggesting that the beverages were added to the diet or displaced energy from foods, not beverages. In this trial, increased reported energy intake and ingestive frequency corresponded with increased body weight with sucrose consumption and the decreased energy intake and ingestive frequency with sucralose consumption corresponded with weight reduction, albeit only as a trend. However, no association between energy intake and saccharin use was observed despite increases in hunger, DTE, and prospective consumption. Whether the increase in body weight among the saccharin group can be attributed to metabolic changes unrelated to energy intake [i.e., changes in metabolic rate (62), gut microbiota energy harvesting efficiency] requires further investigation. Thus, a reduction of energy intake elicited by sucralose with resulting weight reductions remains an open question. A lack of effect of sucralose on appetite or energy intake has been reported previously, but only from short-term feeding trials (15, 20).

Theoretically, if sweetness alone promotes reward-driven feeding, then it may be expected that all sweeteners would exhibit increased DTE and energy intake. However, DTE ratings increased only among the saccharin group and only at selected time points. Further, energy intake did not differ across treatments excluding the energy contributed by sucrose. Sweetness exposure also did not alter the perceived intensity of multiple sensory attributes and overall liking among any treatment group throughout the trial. This suggests that sweetness alone does not account for the weight gain and that additional mechanisms may be involved. Possibilities include effects of LCSs on the CPIR, incretin release, glycemic response, or gut microbiota changes. Elucidation of these mechanisms was not the aim of the trial, but inferences can be drawn based on the observed differences in body weight.

There is speculation that the CPIR can be disrupted with intermittent consumption of nutritive sweeteners and LCSs. This is claimed to dissociate sweetness from the metabolic consequences associated with eating, thereby corrupting a normal control of feeding (18). Saccharin (13) and potentially sucralose (14, 15) are the only LCSs with documented CPIR, yet had significantly different effects on body weight. Thus, this explanation seems unlikely.

An increase in insulin release in the absence of energy due to LCS binding to T1R2/T1R3 receptors on taste and enteroendocrine cells is hypothesized to disrupt glucose homeostasis and increase appetite (6). However, daily sweetener consumption for 12 wk did not alter insulin concentrations during the OGTT in this trial. In addition, LCSs consumed orally or via intragastric administration have no effect on incretin release in humans (63) with the exception of an increase in GLP-1 with consumption of soda sweetened with ace-K and sucralose (64). There, the increase in GLP-1 was attributed to the cola rather than the sweetener and not replicated when the sweeteners were delivered in water (64, 65). Even if sucralose consumption does lead to increased concentrations of GLP-1, the effect of GLP-1 on satiety may be limited, because supra-physiological doses of GLP-1 are necessary to promote satiety (66). The glycemic response during the OGTT was not significantly altered with 12 wk of sucrose consumption despite evidence of increased risk of type 2 diabetes with SSB intake (67). This may be because there was some compensation for the sucrose load resulting in a lower challenge or the duration of exposure was not sufficient for other metabolic changes to manifest. Another RCT that examined the effect of consumption of 1 L/d of SSB for 6 mo among individuals with overweight or obesity also found no effect on glycemic response compared with water or isoenergetic milk controls (Engel, 2017). Still, we cannot conclude that sucrose does not affect glycemia, because effects may just not be observable in trials of only 12-wk or 6-mo duration.

A fraction of saccharin, rebA, and sucralose reaches the colon and has the potential to induce changes in the colonic microbiota that may then increase or decrease energy harvesting efficiency and body weight (22, 68). However, mice fed a high-fat saccharin-supplemented diet or germ-free mice receiving fecal transplantations from saccharin-feed mice experienced no significant changes in body weight or unpurified diet consumption (69). Similar null effects have been reported with sucralose (70, 71) and body weight reduction with aspartame consumption (72). Aspartame is digested in the proximal small intestine, making changes in gut microbial populations difficult to interpret. Changes to the gut microbiota were not monitored in this trial, but the pattern of weight change is not consistent with hypothetical changes based on LCS effects on the microbiota.

This trial has limitations. Because it was designed to address differences between sweeteners, we did not include an unsweetened, water control group. Data from such a group would have helped clarify the effects of fluid intake alone on the study outcomes. Eight participants in the sucrose treatment group withdrew before week 4, so it is unlikely the high attrition rate of the sucrose treatment stemmed from detection of undesired weight gain. In addition, this attrition rate was not observed in the saccharin group who gained a comparable amount of weight. Based on anecdotal comments, we hypothesize that the additional energy from the sucrose beverage somewhat increased satiety (57) and caused greater disruptions to regular meal patterns than the LCS beverages. We are unable to test this hypothesis, because the dropouts occurred before week 4 when dietary recalls and appetite logs were collected. Participants consumed beverage volumes ranging from 1.25 to 1.75 L/d, contributing 400–560 kcal/d. Although these are substantial volumes, mean water from beverages other than plain water is ~1.5 L/d and therefore they fall within a customary intake range (32). Participants were blinded to the treatment and primary aim of measuring body weight in this trial, whereas in the free-living environment, individuals are usually aware of when products contain LCSs. Such knowledge may alter ingestive behaviors (73, 74). EE was measured with tri-axial accelerometers; although they are validated to predict EE and indicated that physical activity was not different in response to the beverage intervention, they provide imperfect measurements of EE. The same holds for diet intake data. Dietary energy is generally underreported, with a greater risk of underreporting among individuals with obesity (75). The critical issue for the present purpose is whether underreporting might be expected to have disproportionately occurred in $\geq$1 treatment groups. There is no evidence of this. LCS concentrations in the beverages were designed to match the sweetness intensity of 8% sucrose. Whereas the intake of all LCSs fell within respective ADI ranges, saccharin and rebA intake exceeded the typical estimated intake of high saccharin and rebA consumers (based on estimates of a 70-kg person) (34, 76, 77). Although LCSs are most commonly consumed as beverages (as in the current trial), they are also increasingly incorporated into food products. Factors such as mastication, food and beverage viscosity, consumption with other nutrients, carbonation, and palatability may all affect ingestive responses to LCSs differently depending on the delivery vehicle. Other LCSs not tested in this trial may have differential effects on body weight. Ace-K, commonly used synergistically with other LCSs, has a similar structure, receptor affinity, and digestibility to saccharin (9, 22, 78). Ace-K was not included in this trial, because it is rarely consumed in isolation. Whether Ace-K consumption leads to similar effects on body weight and whether commercial mixtures of LCSs can be optimized to achieve desired weight management effects warrant testing.

If substantiated through additional testing and confirmed through mechanistic studies, findings from this trial have implications for consumers, clinicians, policy makers, and the food industry. Some LCSs may not hold the anticipated beneficial effects on body weight (e.g., saccharin) and beneficial effects of one LCS (e.g., sucralose) may be attenuated if combined with selected other LCSs. Going forward it will be important to consider each LCS as a distinct entity with respect to its potential health effects.

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